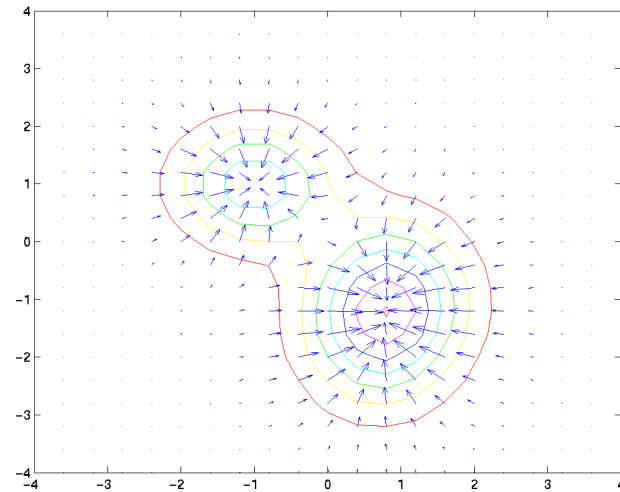
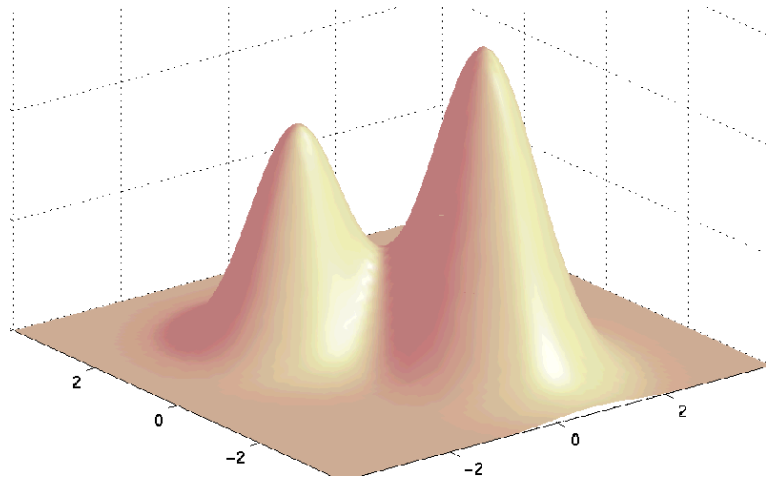


Mathematics of Machine Learning

Optimisation

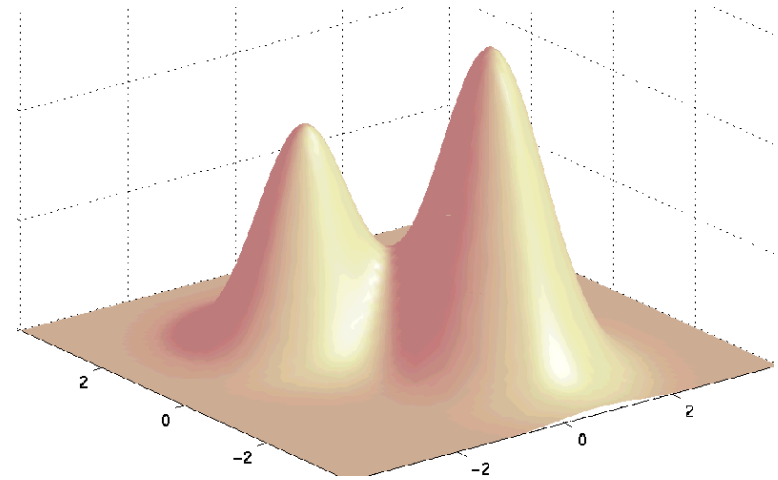


$$z = e^{-(x+1)^2 - (y-1)^2} + 0.6e^{-(x-1)^2 - 0.5(y+1)^2 + 0.1(x-3)(y-3)}$$

Gradient descent, quadratic minima, differing length scales

Outline

1. **Motivation**
2. Gradient Descent
3. Why Gradient Descent is Difficult



ML = Optimisation

- Many learning machines can be thought of as functions of the form

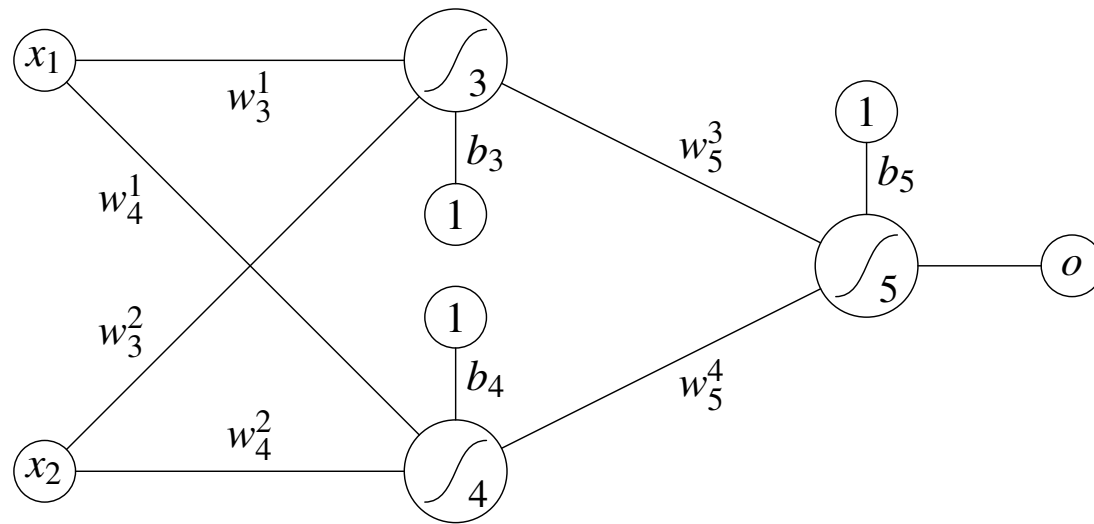
$$\hat{y} = f(x|w)$$

(or more generally $\hat{y} = f(x|w)$)■

- Given an input pattern (set of features) x the learning machine makes a prediction $\hat{y}$ ■
- We try to choose the parameters w so that the predictions are good■
- In practice training a learning machine comes down to optimising some loss function■

MLP

- We can depict a neural network such as an MLP by a diagram



- Stands for the function $(\hat{y} = f(\mathbf{x}|\mathbf{w}))$

$$\hat{y} = g(w_5^3 g(w_3^1 x_1 + w_3^2 x_2 + b_3) + w_5^4 g(w_4^1 x_1 + w_4^2 x_2 + b_4) + b_5)$$

where, for example, $g(V) = \frac{1}{1+e^{-V}}$ ■

Training

- Given a (labelled) training dataset

$$\mathcal{D} = \{(\mathbf{x}_k, y_k) \mid k = 1, \dots, m\}$$

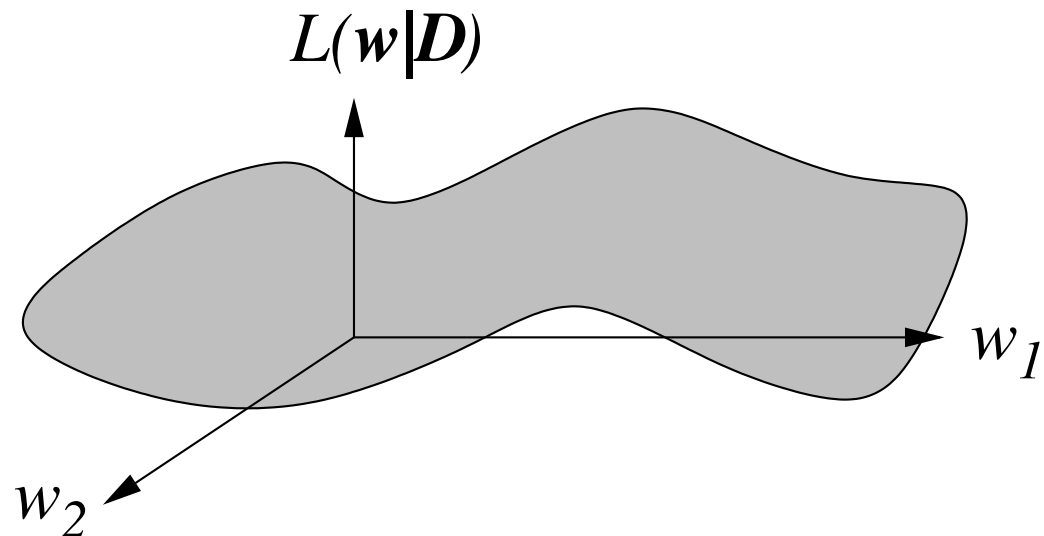
- We define an error or loss function that we want to minimise

$$L(\mathbf{w}|\mathcal{D}) = \frac{1}{m} \sum_{k=1}^m (f(\mathbf{x}_k|\mathbf{w}) - y_k)^2$$

- We then use the machine with the weights $\mathbf{w}^*$ which minimise $L(\mathbf{w}|\mathcal{D})$

Computing Gradients

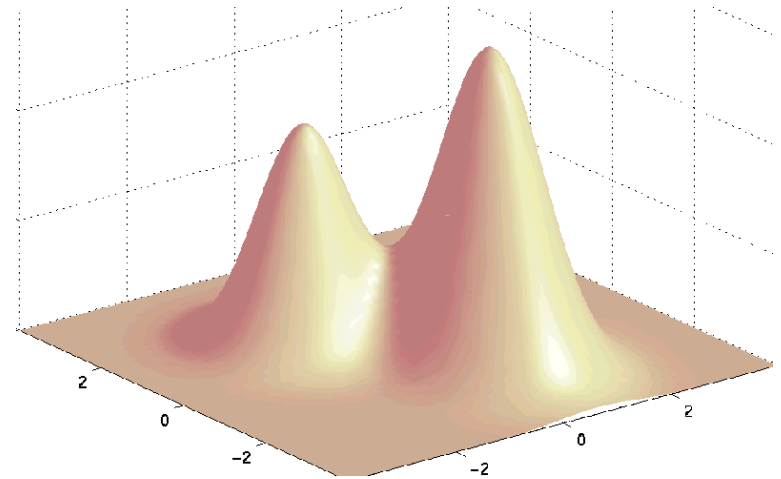
- $L(\mathbf{w}|\mathcal{D})$ is a complex function of the weights $\mathbf{w}$



- To minimise $L(\mathbf{w}|\mathcal{D})$ we compute the gradient $\nabla L(\mathbf{w}|\mathcal{D})$ ■
- For neural networks an efficient algorithm for computing the gradient is known as back-propagation of errors (*back-prop*)■

Outline

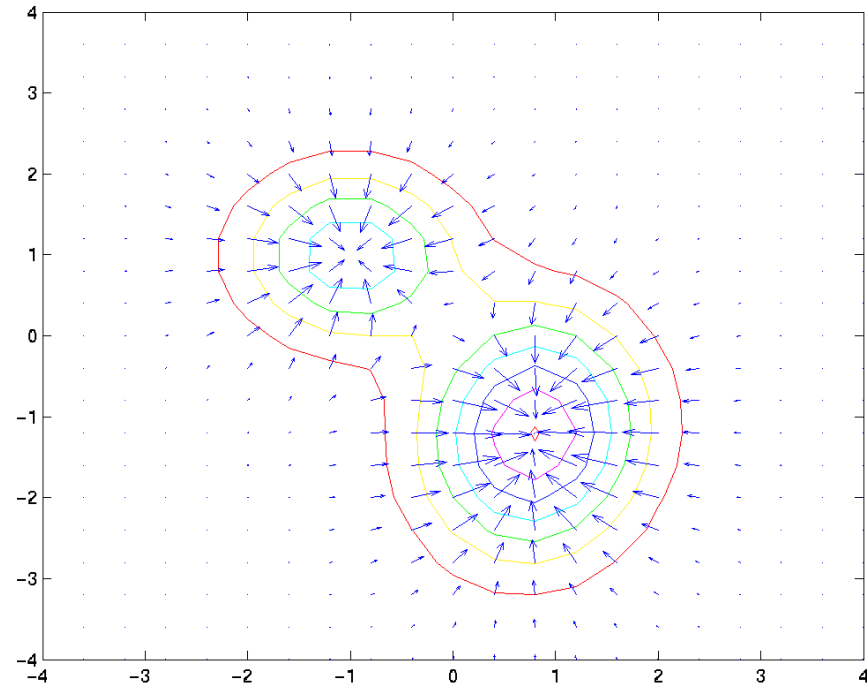
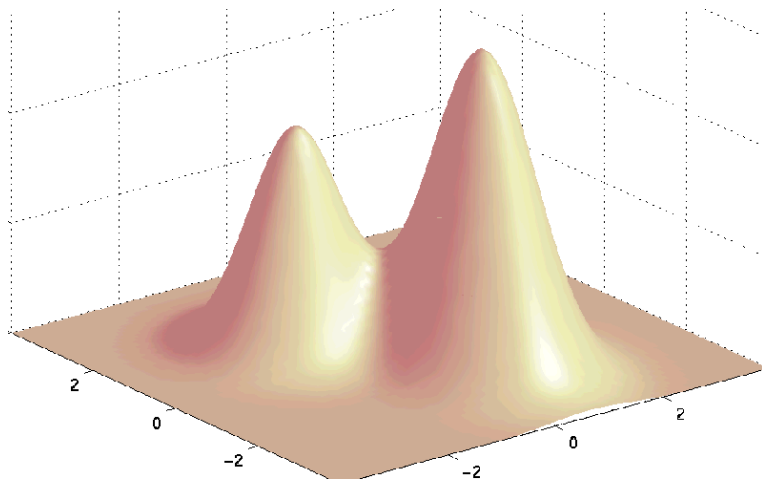
1. Motivation
2. **Gradient Descent**
3. Why Gradient Descent is Difficult



Gradient Optimisation

- At a maximum or minimum (or a saddle point) $\nabla L(\mathbf{w}|\mathcal{D}) = \mathbf{0}$
- E.g.

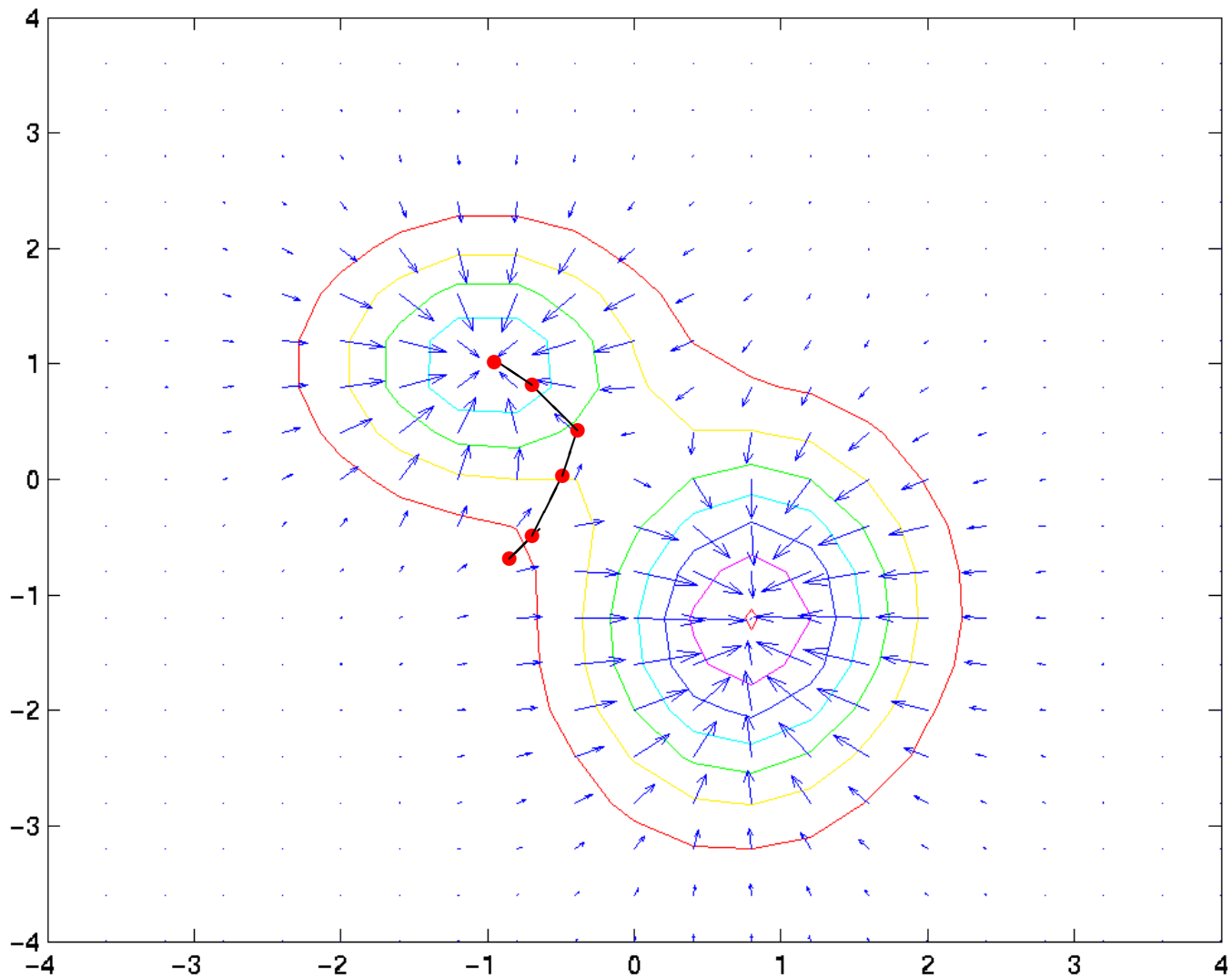
$$L = e^{-(x+1)^2-(y-1)^2} + 0.6e^{-(x-1)^2-0.5(y+1)^2+0.1(x-3)(y-3)}$$



Gradient Descent

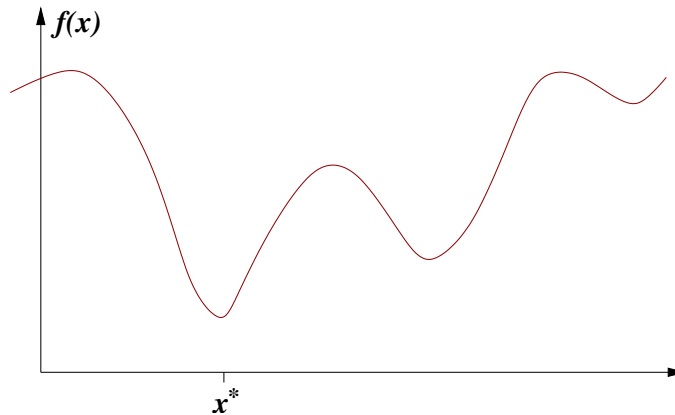
- For a simple function $L(\mathbf{w}|\mathcal{D})$ we can solve $\nabla L(\mathbf{w}|\mathcal{D}) = \mathbf{0}$ explicitly. E.g. the linear perceptron.
- For non-linear functions we usually can't solve this set of simultaneous equations.
- We can find a maximum or minimum **iteratively**.
- If we know the gradient then we can follow the gradient.
 - ★ Maximisation: $\mathbf{w} \rightarrow \mathbf{w}' = \mathbf{w} + r \nabla L(\mathbf{w}|\mathcal{D})$
 - ★ Minimisation: $\mathbf{w} \rightarrow \mathbf{w}' = \mathbf{w} - r \nabla L(\mathbf{w}|\mathcal{D})$

Hill-Climbing



What Goes Right

- Almost all minima are quadratic (Morse's theorem)



- Taylor expanding around a minimum x^*

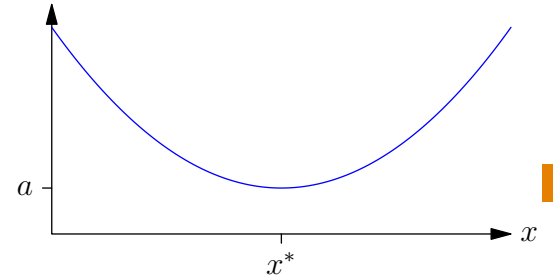
$$\begin{aligned} f(x) &= f(x^*) + (x - x^*) f'(x^*) + \frac{1}{2}(x - x^*)^2 f''(x^*) + \dots \\ &= f(x^*) + \frac{1}{2}(x - x^*)^2 f''(x^*) + \frac{1}{3!}(x - x^*)^3 f'''(x^*) + \dots \end{aligned}$$

- If $x - x^*$ is sufficiently small the higher order terms are negligible

Newton's Method

- If we were in a quadratic minimum

$$f(x) = a + \frac{b}{2}(x - x^*)^2$$



- then

$$f'(x) = b(x - x^*), \quad f''(x) = b$$

- so

$$x - x^* = \frac{f'(x)}{b} = \frac{f'(x)}{f''(x)}$$

- or

$$x^* = x - \frac{f'(x)}{f''(x)}$$

Newton's Method

- This is Newton's method■
- For non-quadratic functions Newton's method converges **quadratically** provided we are sufficiently close to a minimum■
- If we are at a distance $x - x^* = \epsilon$ from the minimum then after one iteration we will be a distance of order ϵ^2 ■, after two iterations of order ϵ^4 , etc.■
- If we are too far from a minimum we might go anywhere!■
- We should follow the gradient until we are near the minimum■

Taylor's Expansion in High Dimensions

- We can generalise these results to many dimensions
- The Taylor expansion of a function $f(\mathbf{x})$ about $\mathbf{x}_0$

$$f(\mathbf{x}) = f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)^\top \nabla f(\mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}_0) + \dots$$

where $\mathbf{H}$ is the **Hessian** matrix with elements

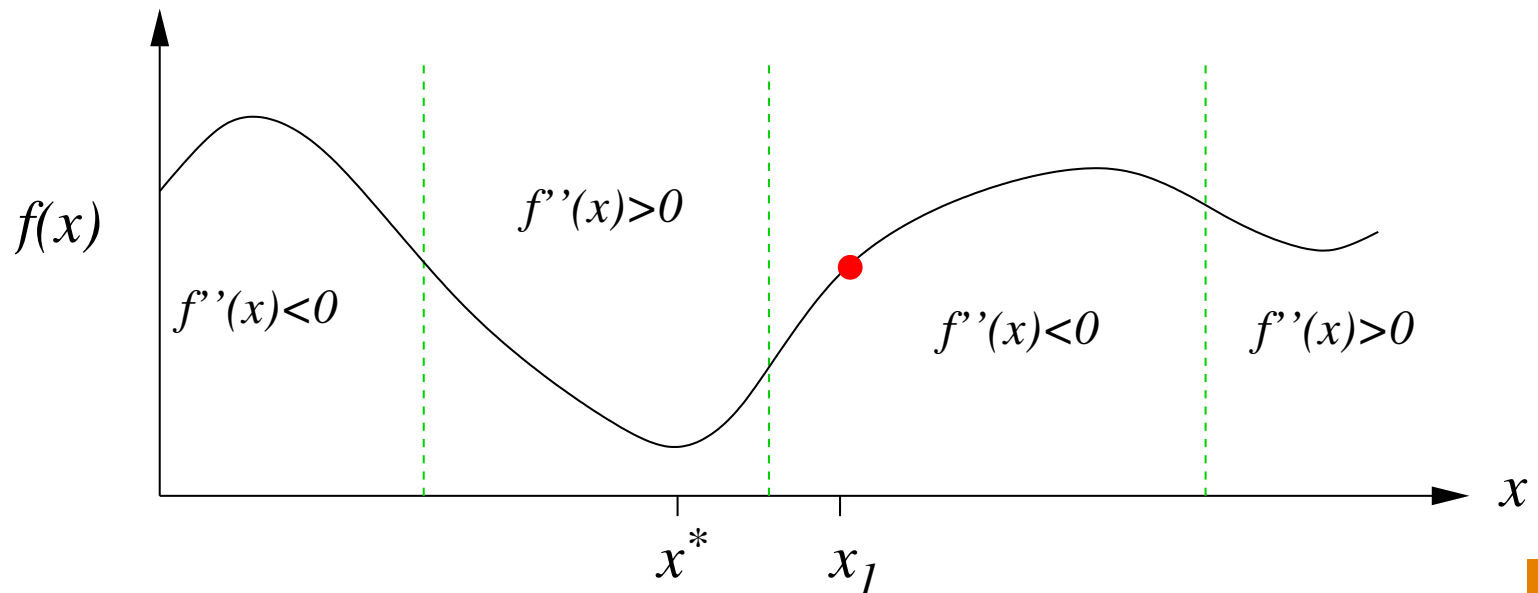
$$H_{ij} = \frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i \partial x_j}$$

- Newton's method in high dimension is

$$\mathbf{x}^* = \mathbf{x} - \mathbf{H}^{-1} \nabla f(\mathbf{x})$$

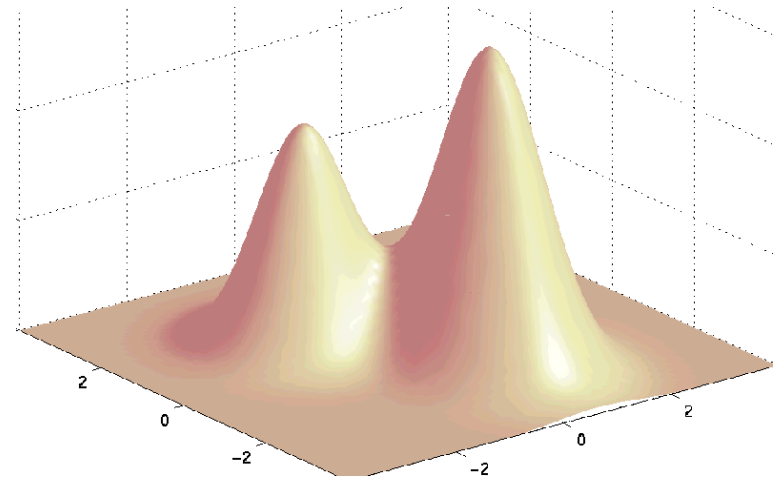
Using the Second Derivative

- If we are optimising N parameters the Hessian is an $N \times N$ matrix
- It is time-consuming to compute (and prone to errors when coding)—for deep learning it is impossible even to store the Hessian
- Away from minima they can be misleading



Outline

1. Motivation
2. Gradient Descent
3. **Why Gradient Descent is Difficult**



Step Size

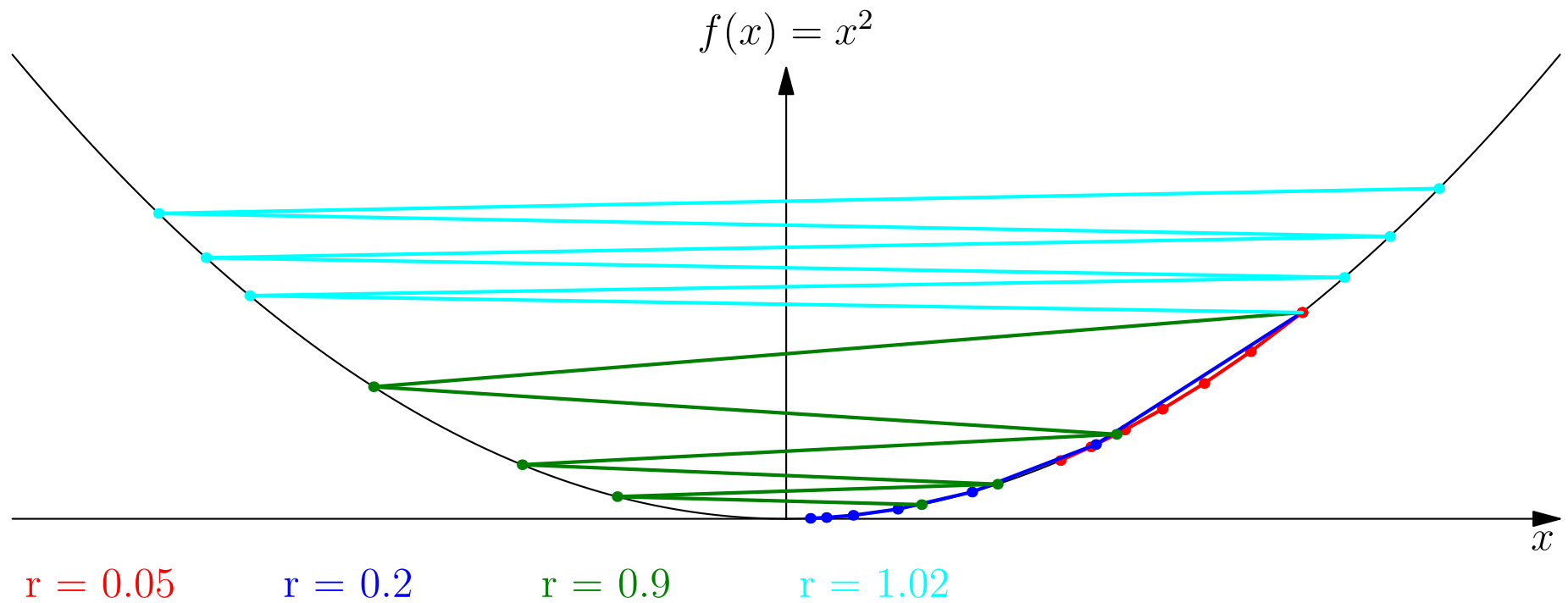
- Gradient descent

$$x' = x - r \nabla f(x)$$

- Need to choose the learning rate or step size, r
- Too small a step takes lots of time
- Too large a step takes you away from a minimum

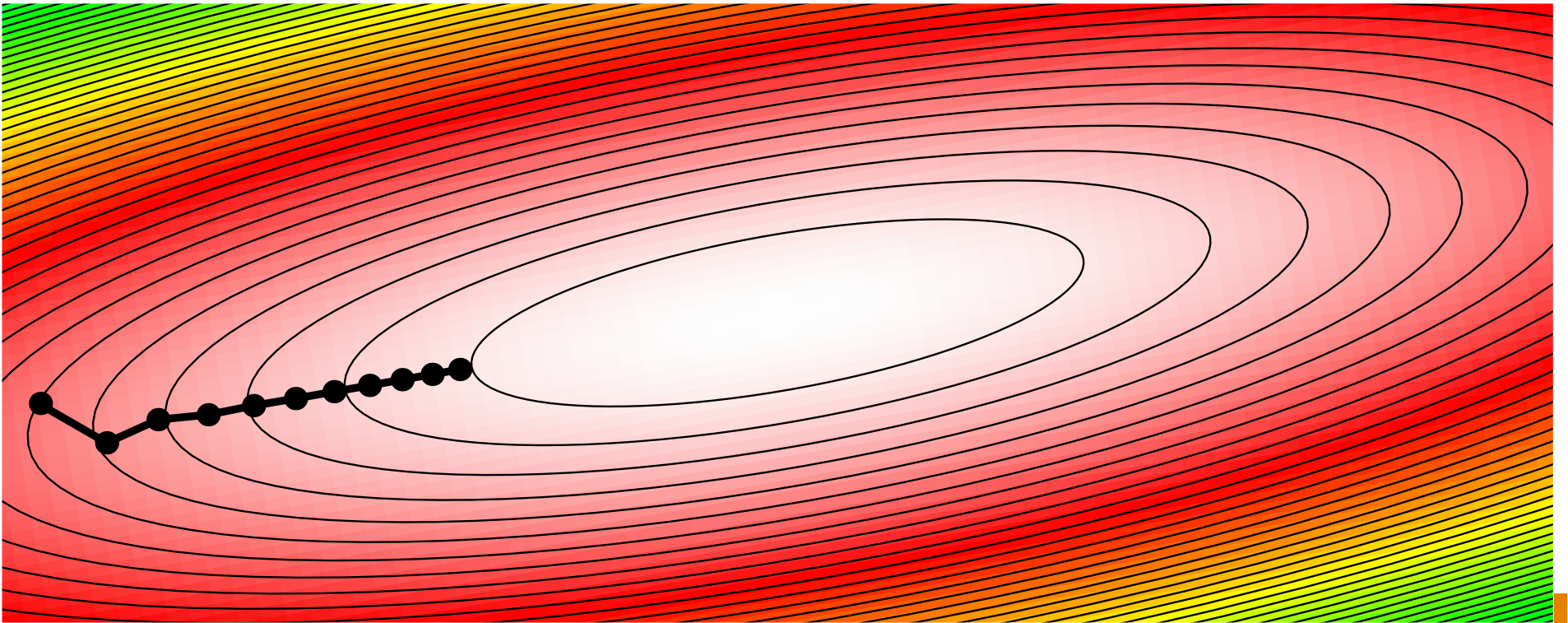
Step Size

$$x \leftarrow x - r f'(x)$$



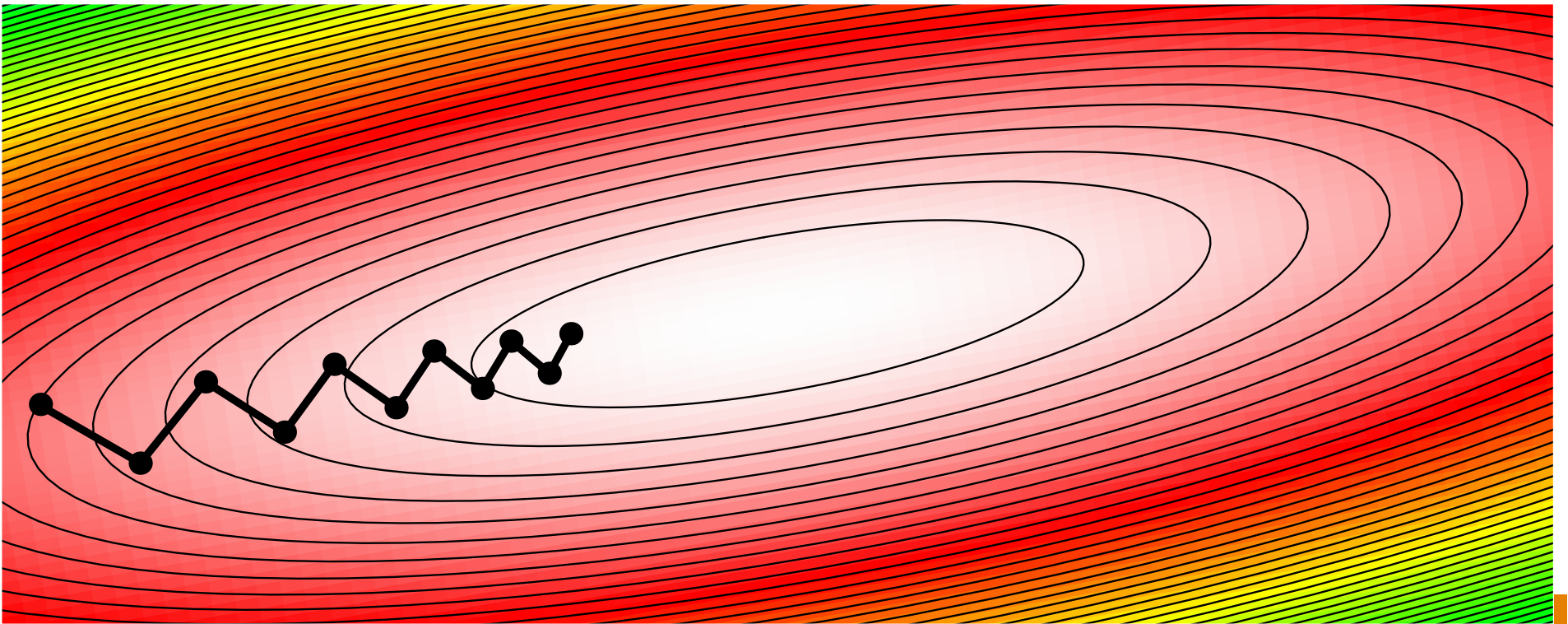
Higher Dimensions

- In higher dimensions the problem is that there are some directions you need to move a long way



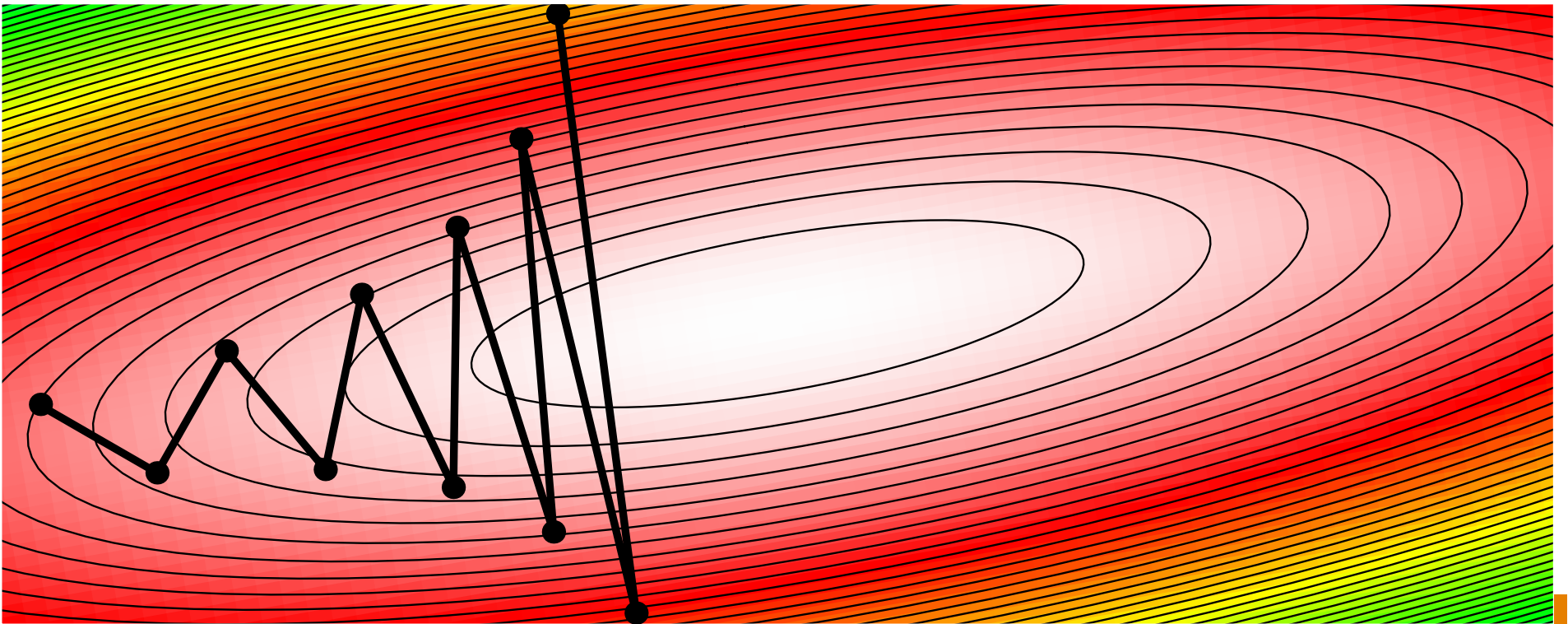
Getting There Quicker

- Increasing the step size speeds up convergence, but the direction of steepest descent doesn't point to the minimum



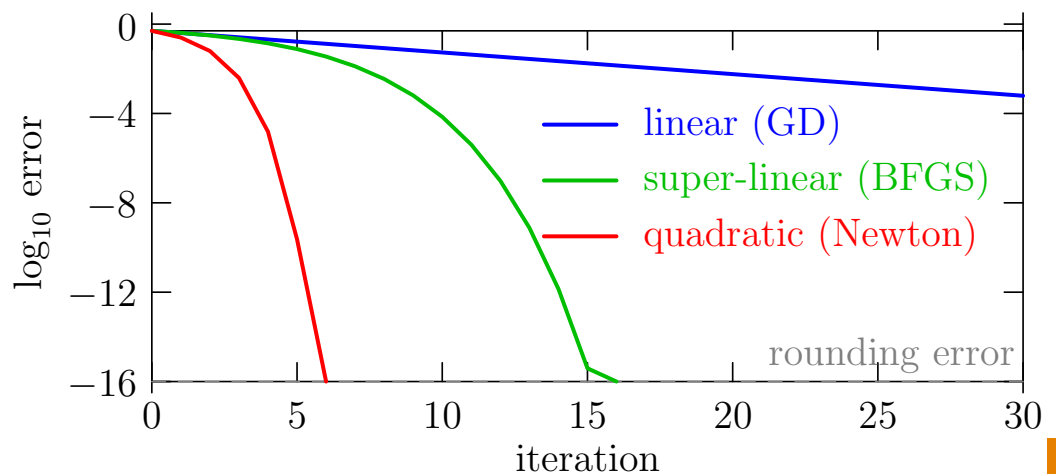
More Haste Less Speed

- Increasing the step size, just a little further, increases the rate of convergence in one direction, but . . . ■



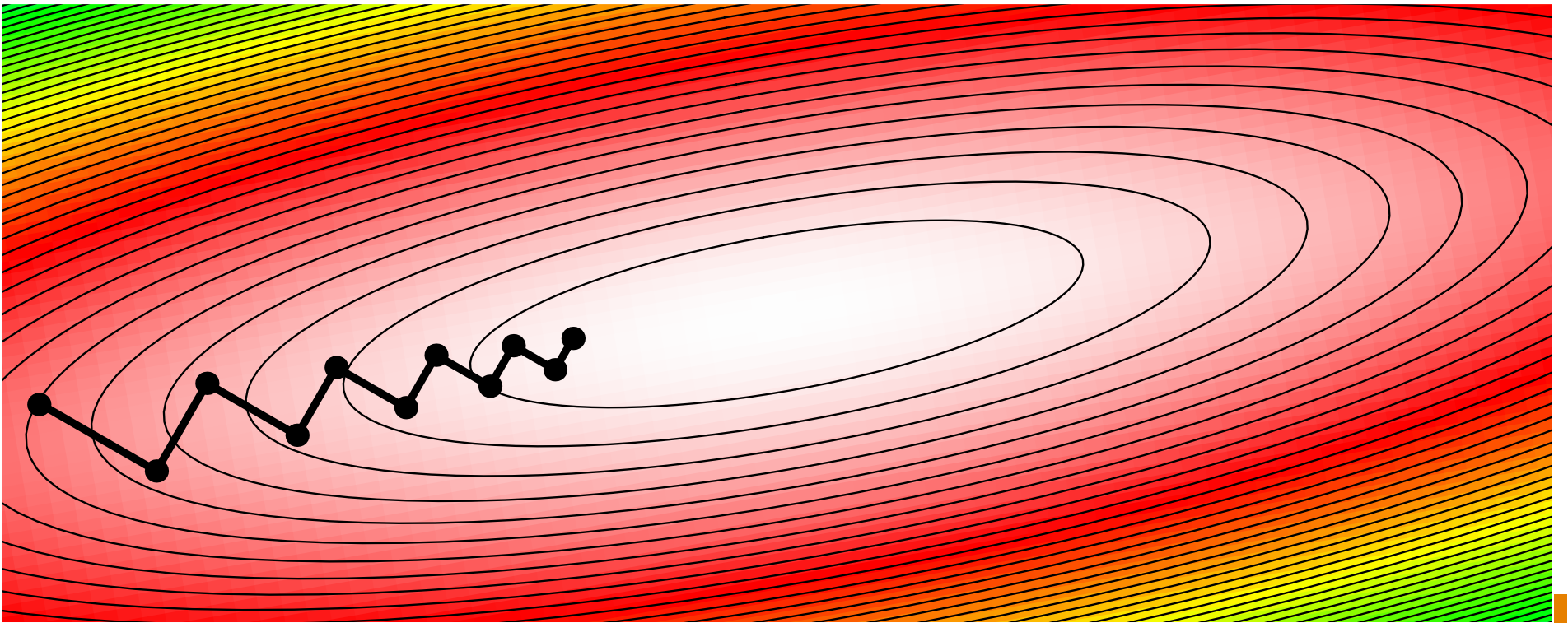
Linear and Quadratic Convergence

- Near a minimum, along each eigenvector of the Hessian, a gradient step multiplies the error by $(1 - r\lambda_i)$ ■
- Stable only if $r < 2/\lambda_{\max}$, so the flattest direction shrinks by only about $1 - \lambda_{\min}/\lambda_{\max} = 1 - 1/\kappa$ per step (κ is the condition number)■
- A constant factor per step is **linear** convergence; Newton squares the error: **quadratic** convergence■



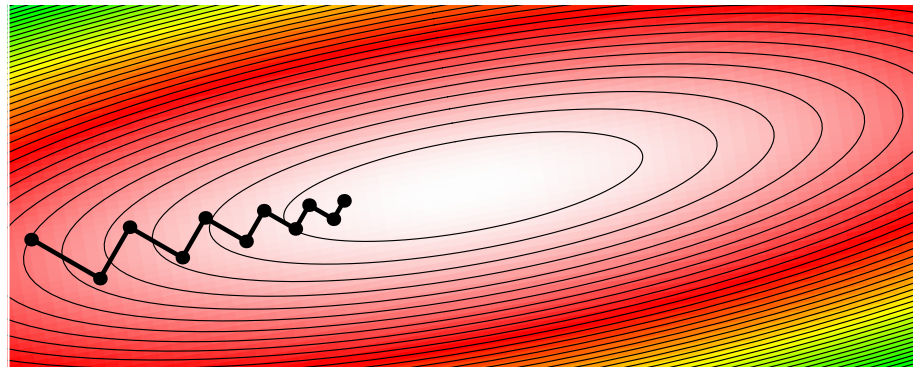
Line Minimisation

- We can systematically seek the minimum along a line of the gradient



Zig-Zag

- Note that in narrow valleys line minimisation zigzags (each step is at right angles to the last)



- If we computed the Hessian and used Newton's method we would jump straight to the minimum if we were in a quadratic potential■
- However computing the Hessian is time consuming and misleading if we are not in a quadratic potential (i.e. far from the optimum)■

Better Optimisation Algorithms

- Good optimisation algorithms use second-order (curvature) information without computing the full Hessian■
- E.g. Conjugate gradient■
 - ★ Performs Line Minimisation■
 - ★ Uses gradient, but does not go along it■
 - ★ For a quadratic minimum in d dimensions it reaches the minimum in d steps■
- E.g. Levenberg-Marquardt■
 - ★ Used on least squares problem only■
 - ★ Uses linear approximation of function to approximate Hessian■
 - ★ Adapts from hill-climbing to Newton method■
 - ★ Avoids line-minimisation■

BFGS and L-BFGS

- BFGS (Broyden–Fletcher–Goldfarb–Shanno) is the most widely used quasi-Newton method■
- It builds an approximation $\mathbf{B} \approx \mathbf{H}^{-1}$ from how the gradient changes from one step to the next

$$\mathbf{B} (\nabla f(\mathbf{x}_{t+1}) - \nabla f(\mathbf{x}_t)) = \mathbf{x}_{t+1} - \mathbf{x}_t \blacksquare$$

- Steps $\mathbf{x}_{t+1} = \mathbf{x}_t - \alpha \mathbf{B} \nabla f(\mathbf{x}_t)$, with a line search for α ■
- Converges **super-linearly** without ever computing the Hessian■
- $\mathbf{B}$ is $N \times N$; **L-BFGS** keeps only the last ten or so steps instead, so its memory is $O(N)$ ■

Levenberg-Marquardt

- Want to minimise $\|\epsilon(\mathbf{w})\|^2$ where $\epsilon_i(\mathbf{w}) = f(\mathbf{x}_i|\mathbf{w}) - y_i$
- Use linear approximation

$$\epsilon_i(\mathbf{w}) \approx \epsilon_i(\mathbf{w}^{(k)}) + (\mathbf{w} - \mathbf{w}^{(k)})^\top \nabla \epsilon_i(\mathbf{w}^{(k)})$$

with $\nabla \epsilon_i(\mathbf{w}^{(k)}) = \nabla f(\mathbf{x}_i|\mathbf{w}^{(k)})$

- Solve quadratic minimisation of approximate error
 $\operatorname{argmin}_{\mathbf{w}} L_{\text{approx}}(\mathbf{w})$ with $\mathbf{J} = \nabla \epsilon(\mathbf{w}^{(k)})$

$$\begin{aligned} L_{\text{approx}}(\mathbf{w}) &= \|\epsilon(\mathbf{w}^{(k)}) + \mathbf{J}(\mathbf{w} - \mathbf{w}^{(k)})\|^2 \\ &= \epsilon(\mathbf{w}^{(k)})^\top \epsilon(\mathbf{w}^{(k)}) + 2(\mathbf{w} - \mathbf{w}^{(k)})^\top \mathbf{J}^\top \epsilon(\mathbf{w}^{(k)}) \\ &\quad + (\mathbf{w} - \mathbf{w}^{(k)})^\top \mathbf{J}^\top \mathbf{J} (\mathbf{w} - \mathbf{w}^{(k)}) \end{aligned}$$

Trust Region

- Solution given by $\nabla_{\mathbf{w}} L_{approx}(\mathbf{w}) = 0$ gives

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - (\mathbf{J}^T \mathbf{J})^{-1} \mathbf{J}^T \boldsymbol{\epsilon}(\mathbf{w}^{(k)})$$

- Can lead us in the wrong direction

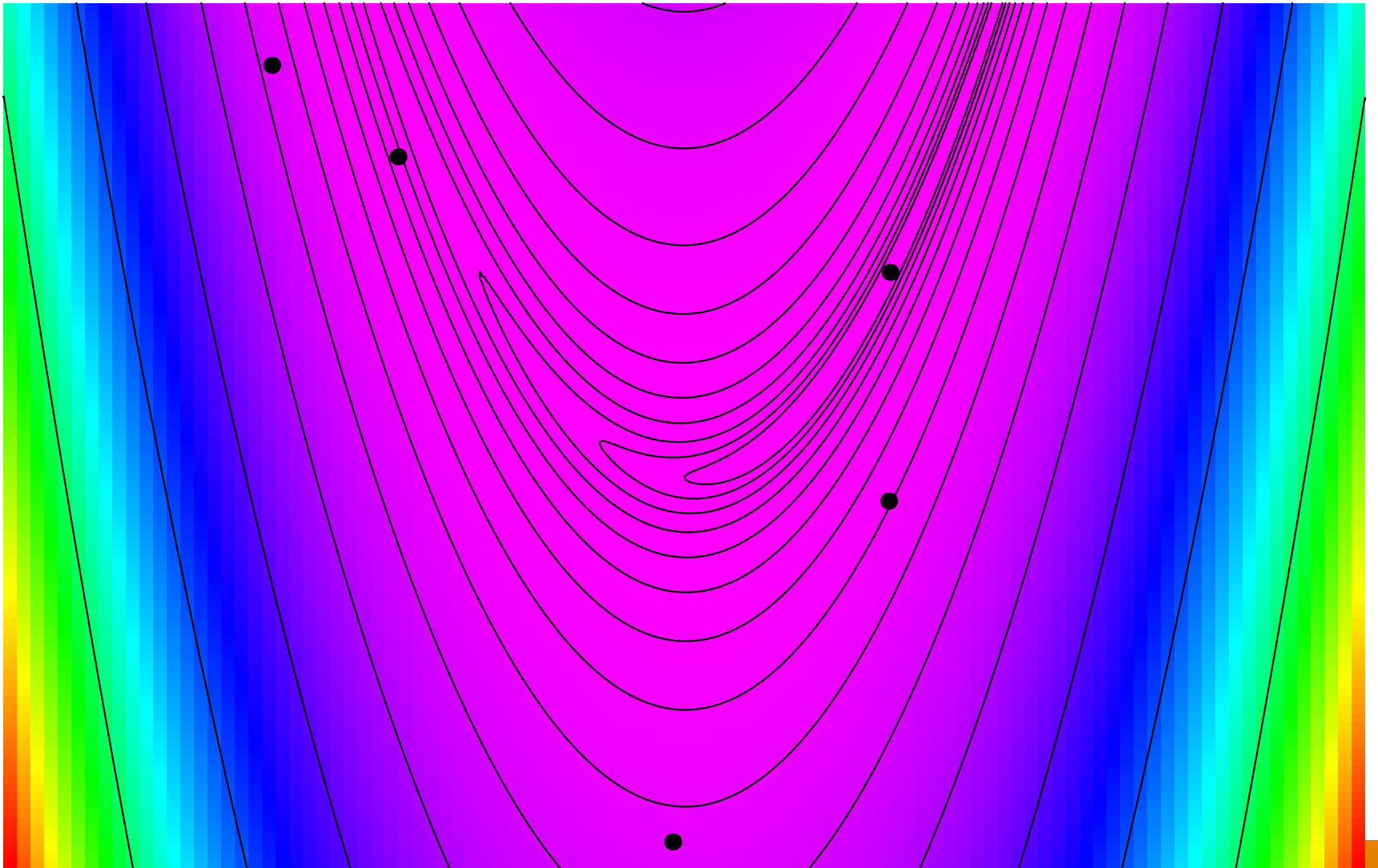
- Instead use $\mathbf{w}^{(k+1)} = \operatorname{argmin}_{\mathbf{w}} L_{approx}(\mathbf{w}) + \nu \|\mathbf{w} - \mathbf{w}^{(k)}\|^2$

$$\mathbf{w}^{(k+1)} = \mathbf{w}^{(k)} - (\mathbf{J}^T \mathbf{J} + \nu \mathbf{I})^{-1} \mathbf{J}^T \boldsymbol{\epsilon}(\mathbf{w}^{(k)})$$

- ν limits the step size

- If predicted reduction in error is accurate then reduce ν , else if predicted reduction in error is very poor increase ν

Rosenbrock: $\epsilon_1 = 10(x_2 - x_1^2)$, $\epsilon_2 = 1 - x_1$



Where Are These Used?

- For up to thousands of parameters (L-BFGS: millions), with gradients computed from all the data■
- **Logistic regression**: scikit-learn's default solver is L-BFGS; Newton's method for generalised linear models■
- **Gaussian processes**: fitting the kernel hyper-parameters with L-BFGS■
- **Curve fitting** (non-linear least squares): Levenberg–Marquardt, e.g. SciPy's `curve_fit`■
- **Support vector machines**: a quadratic programme (later lectures)■
- With millions to billions of parameters and mini-batches of data we need different methods: next lecture■

Summary

- There are some **non-gradient methods** (Nelder Mead, evolutionary strategies, Powell's method), but in very high dimensions these are not very competitive■
- There are **gradient methods** (first order methods) that suffer from the problem of having to choose a single step size with conflicting requirements in different directions; they converge **linearly**■
- **Newton's method** (a second order method) requires computing the Hessian matrix, converges **quadratically**, but can take you in the wrong direction if you are not sufficiently close to a minimum■
- There exist a number of **quasi-Newton methods** (BFGS, conjugate gradient, Levenberg-Marquardt, etc.) that approximate Newton's method often without explicitly computing the Hessian■